

1.5 - Continuity

Goals: Determine where a function is continuous, build continuous piecewise functions, and apply the Intermediate Value Theorem.

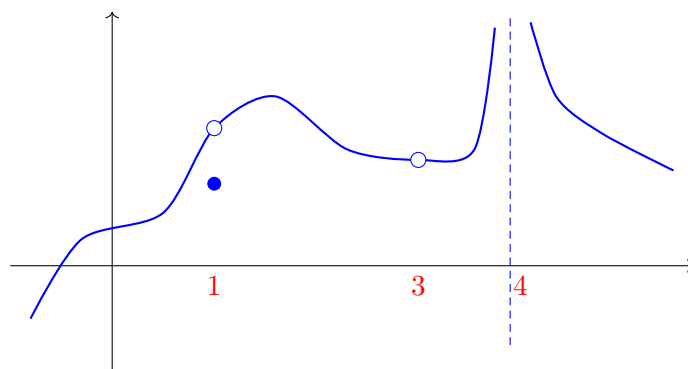
Before Class Videos: Section 1.5 videos 1 and 3

Definition:

A function f is **continuous at** c if all three of the following hold:

1. $f(c)$ exists,
2. $\lim_{x \rightarrow c} f(x)$ exists, and
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

Example 1: Identify the type of discontinuity at each marked x -value.



Example 2: For what values of a is

$$f(x) = \begin{cases} x + 2, & x \leq 1, \\ ax^2, & x > 1 \end{cases}$$

continuous everywhere?

Example 3: For what values of a and b is

$$f(x) = \begin{cases} -2, & x \leq -1, \\ ax - b, & -1 < x < 1, \\ 3, & x \geq 1 \end{cases}$$

continuous everywhere?

Combining Continuous Functions

Theorem:

If f and g are continuous at c , then $f + g$, $f - g$, constant multiples of f , and fg are continuous at c . The quotient f/g is also continuous at c provided $g(c) \neq 0$.

Let's justify the sum rule from the limit laws:

Composition of Functions

Example 4: Compute the domain and range of $\sin \sqrt{x}$

Theorem:

Suppose f is continuous at a and g is continuous at $f(a)$. Then the composition $g \circ f$ is continuous at a , and

$$\lim_{x \rightarrow a} g(f(x)) = g(f(a)).$$

Example 5: On what intervals is $\sin\left(\frac{x-2}{x^2+5x+7}\right)$ continuous?

Example 6: Evaluate

$$\lim_{x \rightarrow 1} \frac{\sin(\pi x/6)e^{x^3} + x}{x^2 + 2}.$$

Theorem:

Intermediate Value Theorem. If f is continuous on $[a, b]$ and y_0 is between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that

$$f(c) = y_0.$$

Sketch:

Example 7: Sketch examples where the assumptions of the Intermediate Value Theorem are not satisfied.

Example 8: Show that

$$x^3 - 3x - 1 = 0$$

has a solution in $[0, 2]$.

Example 9: Use the intermediate value theorem to show that at some time a baby's height equalled their child's height.